

Solution
MAGNETISM -04
Class 10 - Science
Section A

1.
(c) A is true but R is false.
Explanation: A is true but R is false.
2.
(c) A is true but R is false.
Explanation: A is true but R is false.
3.
(c) A is true but R is false.
Explanation: A is true but R is false.
4.
(c) A is true but R is false.
Explanation: Force acting on each pair of the opposite sides of the coil are equal.
5.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
6.
(d) A is false but R is true.
Explanation: A is false but R is true.
7.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: The low electrical resistance of copper makes it a good conductor for electricity.
8.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: In case of the electric field of an electric dipole, the electric lines of force originate from positive charge and, end at negative charge. Since, isolated magnetic lines are closed continuous loops extending throughout the body of magnet, hence they form endless curves.
9.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: In the case of metallic rod, the charge carries flow through the whole of the cross-section. Therefore, the magnetic field exists both inside as well as outside. However, the magnetic field inside the rod will go on decreasing as we go towards the axis.
10.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both the statements are correct, but reason is not the correct explanation of assertion. In testing purposes galvanometer is used to measure polarity. There is a torque acting on a current carrying coil suspended in a magnetic field which produces deflection.
Deflection is directly proportional to current flowing through the galvanometer coil.
11.
(c) A is true but R is false.
Explanation: Assertion is correct but the reason is wrong. As the current changes in the conductor, magnitude of the magnetic field produced also varies which is the reason for a change in the deflection of the iron filings.
12.
(a) Both A and R are true and R is the correct explanation of A.
Explanation: The magnitude of the magnetic field is
 - i. Directly proportional to the current I passing through the wire.
 - ii. Inversely proportional to the distance r from the wire. The magnetic field is stronger at a point that is nearer to the conductor and goes on decreasing on moving away from the conductor.

13. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Magnetic force is always perpendicular to the direction of motion of charged particle, i.e., work done on the charge particle moving on a circular path in magnetic field zero.
14. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
15. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The magnetic field lines form closed loops unlike electrostatic field lines which originate from the charge and end at charge. Mono-magnetic pole does not exist in nature.
16. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both the statements are correct individually, but the reason is not the correct explanation of assertion. As when a fault occurs in the domestic lines there is a protective device called fuse which breaks the circuit and protects all the equipments by melting the low melting point wire wrapped to it in order to interrupt supply.
17. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: If the wires are twisted together, they can be modelled as a single wire carrying current in the opposite directions. In this model, no magnetic field is induced in the wires which does not affect adjacent circuits.
18. **(c)** A is true but R is false.
Explanation: It is used to find the direction of force in a current-carrying conductor in the presence of magnetic field.
19. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: An electric current is equivalent to the charges (or electrons) in motion. Such charges produce magnetic interaction. The magnetic field produced by current interacts with magnetic needle and deflects it.
20. **(d)** A is false but R is true.
Explanation: When a charged particle moves perpendicular to magnetic field, it experiences a force which changes the direction of motion of the particle without changing the magnitude of velocity of the particle. Hence kinetic energy remains constant but momentum of electron changes.
21. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: A force is always exerted due to magnetic field in the same way electric current flowing through any conductor produces magnetic field. And in this case, Fleming's left-hand rule is used to predict directions of the magnetic field, current and displacement.
 So, assertion and reason both are true and reason is the correct explanation of assertion.
22. **(d)** A is false but R is true.
Explanation: A is false but R is true.
23. **(d)** A is false but R is true.
Explanation: A is false but R is true.
24. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
25. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: The torque acting on a coil is given by,

$$\tau = NIA \sin\theta$$
 where θ is the angle between the plane of the coil and the direction of magnetic field. When $\theta = 90^\circ$, then $\tau = 0$. The coil tries to orient itself in this position. Thus in equilibrium, the coil acquires a position, such that its plane makes an angle 90° with the direction of magnetic field.
26. **(c)** A is true but R is false.

Explanation: The magnitude of magnetic field produced by a current carrying circular coil is maximum at the centre and is not proportional to the distance of a point from the circular coil.

27. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
28. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: The metallic body of the electrical appliance is connected to the third pin which is connected to the earth. This is a safety precaution and avoids eventual electric shock. By doing this the extra charge flowing through the metallic body is passed to earth and avoid shocks. There is nothing such as reducing the heating of connecting wires by three-pin connections.
29. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: The magnetic field is independent of length and area. It is uniform inside the solenoid.
30. (c) A is true but R is false.
Explanation: From equation $F = qvB \sin\theta$
Force on moving charge will be maximum if the direction of the velocity of charge is perpendicular to direction of magnetic field (when $\theta = 90^\circ$)
31. (a) Both A and R are true and R is the correct explanation of A.
Explanation: In a rectangular coil carrying a steady current, the direction of current in opposite sides of coil is opposite to each other, therefore, forces acting on each pair of the opposite sides of the coil are equal and opposite, i.e., net force on the coil is equal to zero.
32. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: The wires are parallel to each other but the direction of current in it is in same direction so they attract each other. If the current in the wire is in opposite direction then wires repel each other.
33. (a) Both A and R are true and R is the correct explanation of A.
Explanation: The magnetic field at the centre of circular coil is directly proportional to the current flowing through it. So if current through coil is doubled then magnetic field becomes double.
34. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both statements are true but logic does not support either. The conductor experiences force as a result of the magnetic field's interaction with the external magnetic field.
35. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
36. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
37. (c) A is true but R is false.
Explanation: When we use high voltages for transmission system then line losses reduces to a much extent. As high voltages are used so current will be less in those cases which in turn reduces I^2R losses occurring in the transmission lines. So, efficiency of the transmission lines increases. Thus, assertion is true but reason is false.
38. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
39. (c) A is true but R is false.
Explanation: In a conductor, the average velocity of electrons is zero. Hence no current flows through the conductor. Hence, no force acts on this conductor.

40. (c) A is true but R is false.
Explanation: According to Fleming's left-hand rule, the direction of the magnetic force on a moving charge is always perpendicular to the magnetic field.
41. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Here while carrying out an experiment if a compass needle is placed near a wire carrying current then due to the effect of magnetism which is produced due to electric current produced in the wire the needle gets deflected. Which shows that the magnetism and electricity are interlinked. So, both assertion and reason are true and reason is the correct explanation of assertion.
42. (c) A is true but R is false.
Explanation: Magnetic field is not a scalar quantity rather it is a vector quantity which has both magnitude and direction. So, when iron filings are placed around a magnet, they get arranged in a fashion similar to that of magnetic lines. So, assertion is correct, but reason is false.
43. (d) A is false but R is true.
Explanation: A is false but R is true.
44. (c) A is true but R is false.
Explanation: A is true but R is false.
45. (d) A is false but R is true.
Explanation: When current flows through a solenoid, the currents in the various turns of the solenoid are parallel and in the same direction. Since the current flowing through parallel wires in the same direction lead to force of attraction between them, the turns of the solenoid will also attract each other and as a result the solenoid tends to contract.
46. (c) A is true but R is false.
Explanation: A is true but R is false.
47. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: When a current-carrying conductor is placed in a magnetic field, it experiences a force except when it is placed parallel to the magnetic field. The force acting on a current-carrying conductor in a magnetic field is due to interaction between magnetic field produced by the current-carrying conductor and external magnetic field in which the conductor is placed.
48. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.

Section B

49. Fill in the blanks:

- (i) 1. Electromagnet
- (ii) 1. Curved
- (iii) 1. Overloading
- (iv) 1. Earth wire
- (v) 1. Fuse
- (vi) 1. Fuse
- (vii) 1. Green
- (viii) 1. Parallel
- (ix) 1. Red
- (x) 1. Solenoid
- (xi) 1. Compass needle
- (xii) 1. Closed

Section C

50. **(c)** 1-B, 2-D, 3-A, 4-C
Explanation: (1) The tesla (symbolized T) is the standard unit of magnetic flux density.
(2) A solenoid is a tightly wound helical coil of wire whose diameter is small compared to its length. The magnetic field generated in the centre, or core, of a current carrying solenoid is essentially uniform, and is directed along the axis of the solenoid.
(3) The needle of a magnetic compass is 'just' a small bar magnet that is balanced carefully so it can rotate freely.
(4) Solenoid is temporary strong magnet, when connected through external source like battery or when electricity passed through it. It is also known as electromagnet.
51. **(d)** 1-B, 2-D, 3-A, 4-C
Explanation: (1) Michael Faraday is credited with the discovery of electromagnetic induction on August 29, 1831.
(2) A current-carrying conductor, when placed in a magnetic field, experiences a force. If the direction of magnetic field and that of current are mutually perpendicular then force acting on the conductor will be perpendicular to both and will be the given by Fleming's left-hand rule.
(3) The direction of the magnetic field lines of force around a conductor is given by the Maxwell's right hand grip rule or the right handed corkscrew rule.
(4) Hans Christian Oersted discovered Electromagnetism.
52. **(c)** (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)
Explanation:
 - Green wire stands for grounding/ earthing in the electric circuit.
 - A dry cell is a type of battery. It always produces a DC current.
 - A solenoid is a coil wound into a tightly packed helix.
 - An electromagnet is a temporary magnet that has a magnetic field produced by the flow of electric current in a coil wound around a soft iron core.
53. **(a)** 1-A, 2-C, 3-B, 4-D
Explanation:
 - i. A lodestone is a naturally magnetized piece of the mineral magnetite. Loadstone is iron ore.
 - ii. Magnetic flux is the number of magnetic field lines passing through a surface placed in a magnetic field.
 - iii. An electric motor converts electrical energy into mechanical energy.
 - iv. Permanent magnet are usually made of hard materials which are strongly magnetized. A perfect example of permanent magnet is the Bar Magnet. In fact, Permanent magnets are most of the times referred to as Bar Magnets also.
54. **(c)** (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)
Explanation:
 - Dynamo is a device that makes direct current electric power using electromagnetism. It is also known as a generator. Dynamos and generators work using the wild complex phenomena of electromagnetism.
 - Magnetism is one aspect of the combined electromagnetic force. It refers to physical phenomena arising from the force caused by magnets, objects that produce fields that attract or repel other objects.
 - A generator converts mechanical energy into electrical energy, while a motor converts electrical energy into mechanical energy. Both the devices work because of electromagnetic induction, which is when a voltage is induced by a changing magnetic field.
 - Andre-Marie Ampere was a French physicist and mathematician who was one of the founders of the science of classical electromagnetism. The SI unit of measurement of electric current, the ampere, is named after him.
55. **(d)** (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
Explanation:

- Like poles repel each other.
- Unlike poles attract each other.
- Live wire is red in colour.
- The neutral wire is black in colour.

Section D

56.

(b) 17°

Explanation: Magnetic axis is the line passing through earth's magnetic north and south poles which is inclined at an angle of nearly 17° with the geographic axis.

57. (a) Straight

Explanation: The magnetic field lines inside the solenoid are in the form of parallel straight lines. The field lines inside the solenoid are in the form of straight lines which indicates that the magnetic field is the same at all points inside the solenoid.

58.

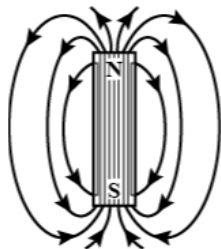
(c) Live wire and Neutral wire

Explanation: When the main switch in a building is turned off or the breaker trips or the fuse burns, power to both live and neutral lines is disconnected. The earth or ground line is a safety line that must never be disconnected.

Section E

59. i. No two magnetic field lines are found to cross each other. If two field lines crossed each other, it would mean that at the point of intersection, the compass needle would point in two directions at the same time, which is not possible.
- ii. The magnetic field and hence the magnetic line of force exist in all the planes all around the magnet.
- iii. The relative strength of the magnetic field is shown by the degree of closeness of the field lines and the direction of the magnetic field is obtained by tangent to the field lines at the point of intersect.

OR



60. i. Magnetic field inside infinite solenoid is uniform. Hence it is same at all points.
- ii. Clock face rule
- iii. For a long solenoid, magnetic field $B \propto In$; where I is the flowing current and n is number of turns per unit length in the solenoid. Therefore, in the given case magnetic field will remain unchanged.

OR

N-pole

61. i. The deflection in galvanometer can be seen if bar magnet moved towards or away from coil parallel to the axis of the coil.
- ii. If the needle of the galvanometer deflects it means there is change in magnetic field and current is induced.
- iii. By Faraday's law of electromagnetic induction, the e.m.f. induced in a conductor is proportional to the rate of change of magnetic lines of force linking the circuit. Hence, by pushing in the magnet faster, the rate of change of magnetic lines will increase. This results in a larger induced e.m.f. and hence, larger deflection of the meter.

OR

The number of turns of the coil, The speed with which the magnet is moved, and the strength of the magnet.

62. i. Fleming's left hand rule is used to determine the direction of force on electron i.e., in south direction.
- ii. The angle between velocity and magnetic field is zero. Therefore, magnetic force on the particle is zero.
- iii. As the direction of current is taken opposite to the direction of motion of electrons, therefore, current from the motion of electron and proton is in the same direction, i.e., from bottom to top. Now, according to Fleming's left hand rule, the electron and the proton experience forces both pointing into the plane of paper.

OR

into the page